Secondary Containment System

304B Chapter 4.2

Objectives

1. Identify the system's purposes.
2. Explain the function and operation of the following secondary containment features:
   a. Reactor building
   b. Relief panels
   c. Airlocks
   d. Reactor building normal ventilation system isolation valves and dampers
   e. Unit area coolers
   f. Primary containment purge system

Objectives

3. List the conditions that will isolate the reactor building normal ventilation system (RBNVS).
4. Explain the following flow paths:
    a. Air transfer through the RBNVS
    b. Purge vent & exhaust from primary containment
5. Define secondary containment integrity.

Objectives

6. Summarize the response of secondary containment equipment response to accident conditions.
7. Describe how secondary containment interrelates to the following:
   a. Primary Containment System
   b. Reactor Building Standby Ventilation System
   c. Nuclear Steam Supply Shutoff System

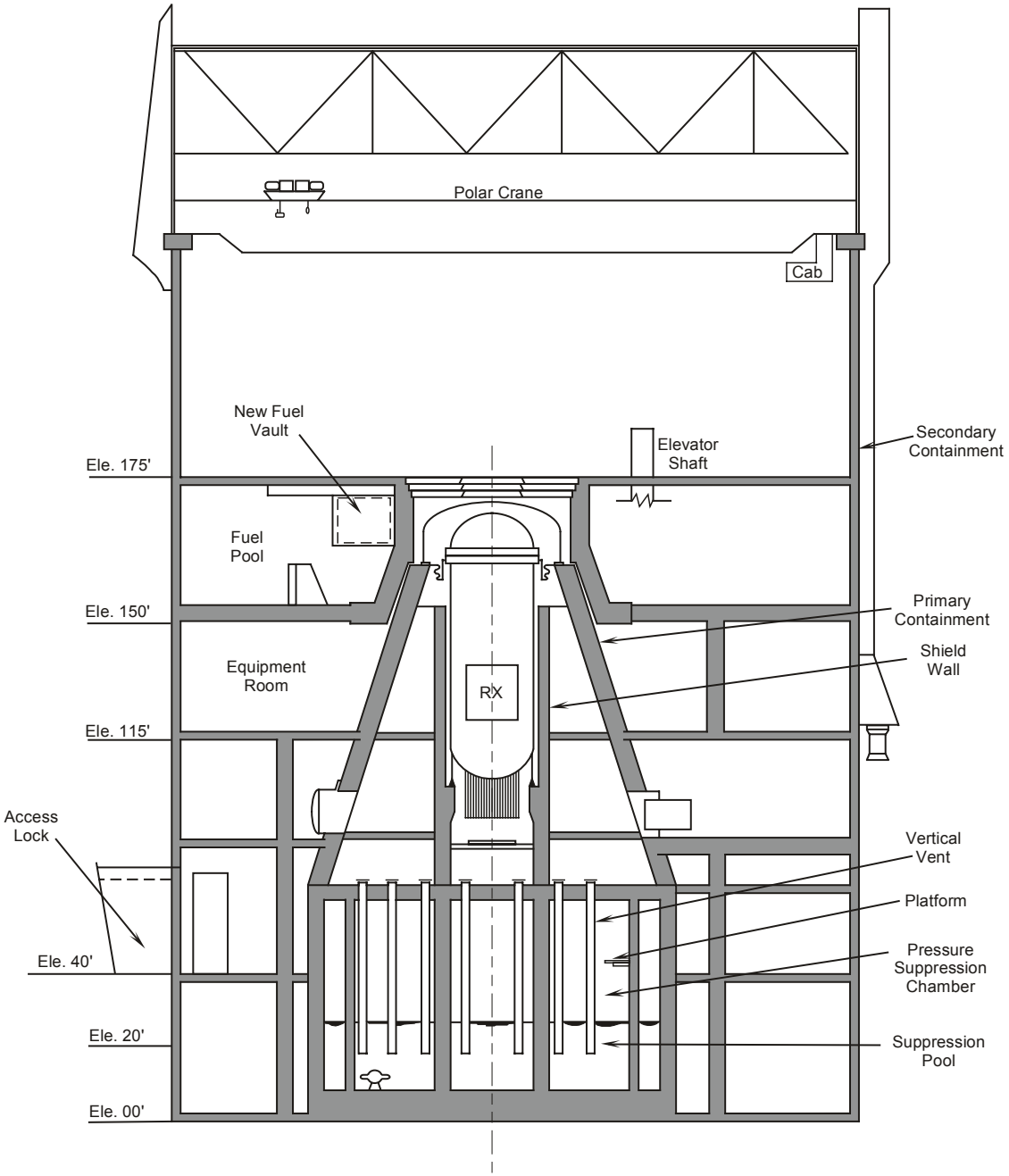


Figure 4.2-1 Mark II Containment

- Reactor building
- Relief panels
- Air locks
- Reactor building normal ventilation system (RBNVS)
- Primary containment purge system (PCPS)

Purpose

- The purposes of the Secondary Containment System are:
  - minimize the ground level release of radioactive material following an accident
  - to provide primary containment when the drywell or suppression chamber are open.

Reactor Building

- Concrete
- Blowout panels
- Encloses primary containment
- Serves as primary containment when drywell or suppression chamber are open

Relief panels

- turbine building blowout panels
- main steam tunnel relief vents
- the reactor building exterior metal siding.

Airlocks

Entry to and exit from the reactor building are through double door personnel and equipment air locks. Each pair of access doors is equipped with rubber weather-strip type seals. The doors are electrically interlocked so that only one of the pair may be opened at a time.

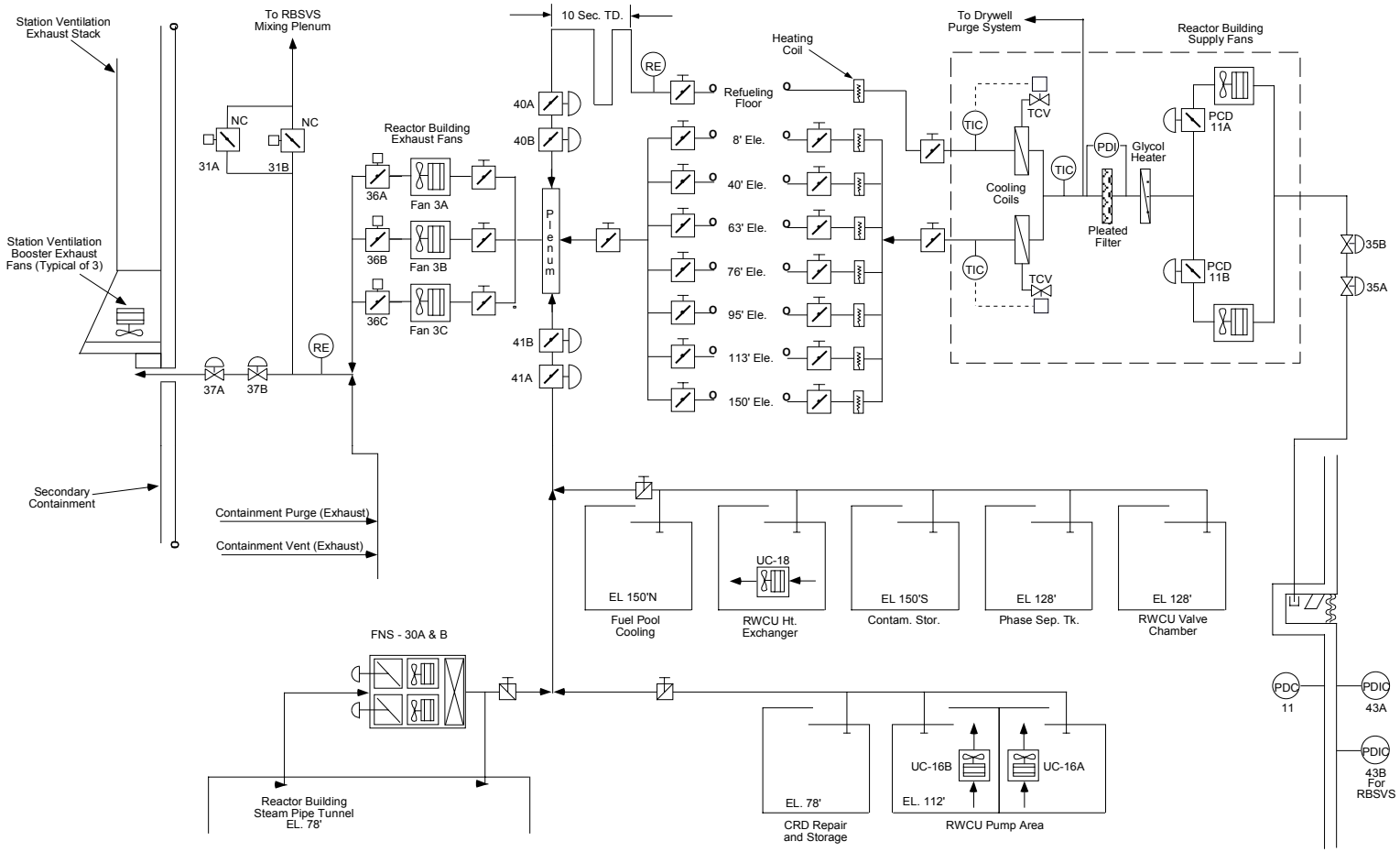


Figure 4.2.2 Reactor Building Normal Ventilation System

0396

RBNVS

The RBNVS consists of:

- heating and air conditioning supply units with two supply fans
- three exhaust fans
- distribution ducts
- manually positioned ventilation louvers
- differential pressure controlled dampers
- unit coolers
- controls and instrumentation.

Sub-Atmospheric Pressure

- - 1.5 inch ΔP
- Prevents out-leakage
- Pressure control dampers

RBNVS Supply Flow

- Intake louver
- Supply fans
- Pressure Control Dampers
- Heaters & coolers
- Distribution ducts

RBNVS Exhaust Flow

- Exhaust plenum
- Refuel area
- Exhaust fans
- Station ventilation exhaust duct

Area Unit Coolers

- Local
- Some are safety-related
- Auto-initiation
- RBSVS/Control Room Air Conditioning Chilled Water System cooling water

Primary Containment Purge

- Primary Containment Purge System (PCPS)
  - a high efficiency particulate absolute (HEPA) filter
  - a charcoal adsorber
  - a fan
- De-inerts primary containment to allow normal personnel access

Secondary Containment Integrity

- The operational conditions that require that secondary containment integrity must be maintained include:
- all times that primary containment integrity is required (i.e.; the reactor is critical or moderator temperature is >212F and fuel is in the reactor vessel)
- when fuel handling operations are in progress within the secondary containment
- when activities are being performed that have a potential for draining the reactor vessel.

Secondary Containment Integrity

- reactor building is intact
- RBNVS automatic isolation valves are operable or deactivated in the closed position.
- RBSVS is operable.
- Secondary containment pressure is less than or equal to -1.5 inch H2O gauge.
- Secondary containment leakage rates are within specified limits.
- All containment penetration seal mechanisms are operable.
- At least one door in each access opening is closed.
- All equipment hatches are closed and sealed.

Operations

- Normal
- Infrequent
- System Isolation

Isolation Signals

- High drywell pressure high (≥1.69 psig)
- Reactor vessel low low level (Level 2, -38 inches)
- Refuel area exhaust ventilation radiation high (≥35 mR/hr)
- Vent/purge exhaust ventilation radiation high (≥5.4 x 105 cpm)
- Reactor Building low ΔP (> -0.30" H2O)
- RBSVS initiation signal, multiple inputs.

System Interfaces

- **Primary Containment System (Chapter 4.1)**
- **Reactor Building Standby Ventilation System (Chapter 4.3)**
- **Nuclear Steam Supply Shutoff System (Chapter 4.4)**
- **Emergency AC Power (Chapter 9.2)**

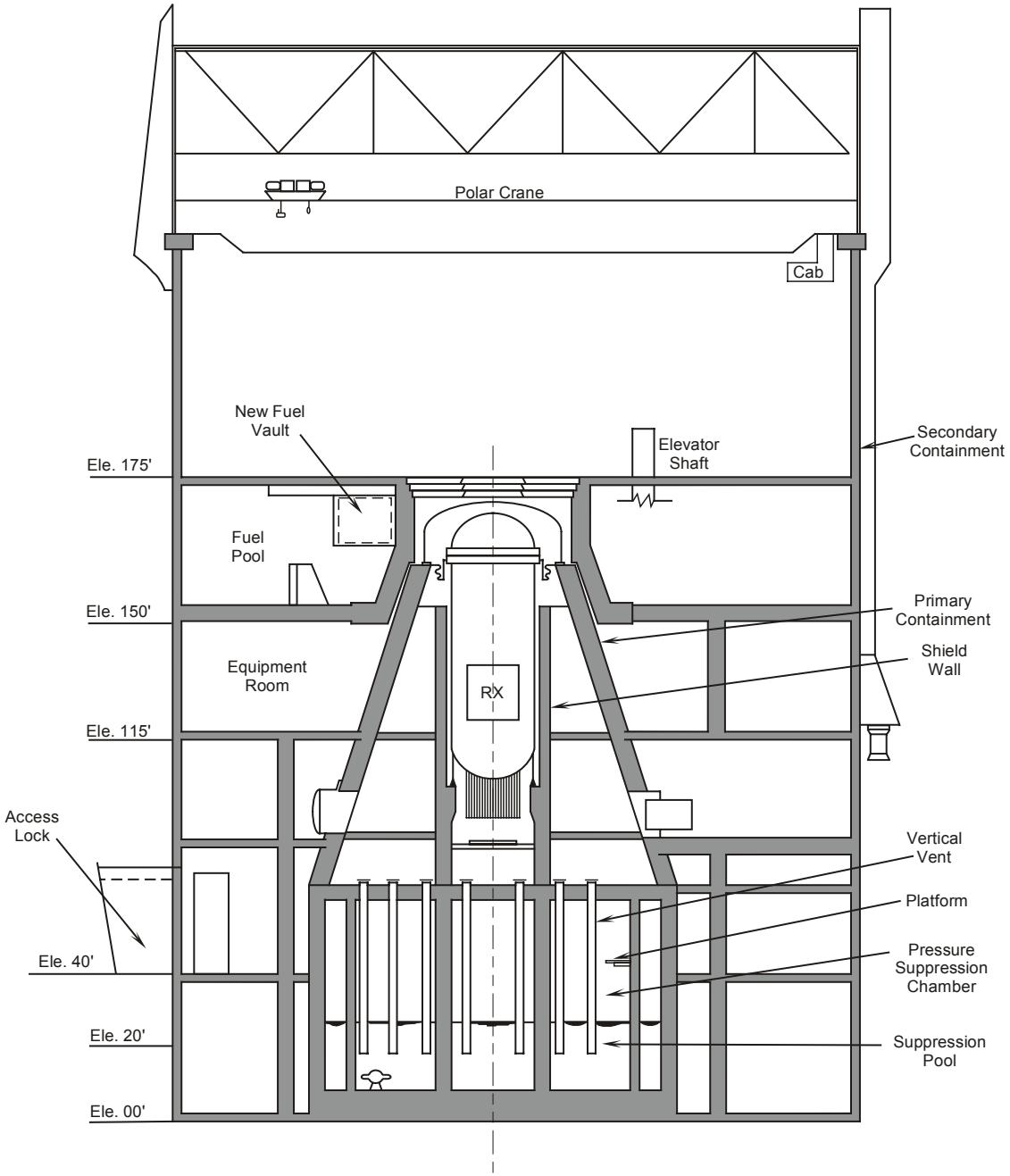


Figure 4.2-1 Mark II Containment

REVIEW

- Relief panels
- Air locks
- Reactor building normal ventilation system (RBNVS)
- Primary containment purge system (PCPS)

REVIEW

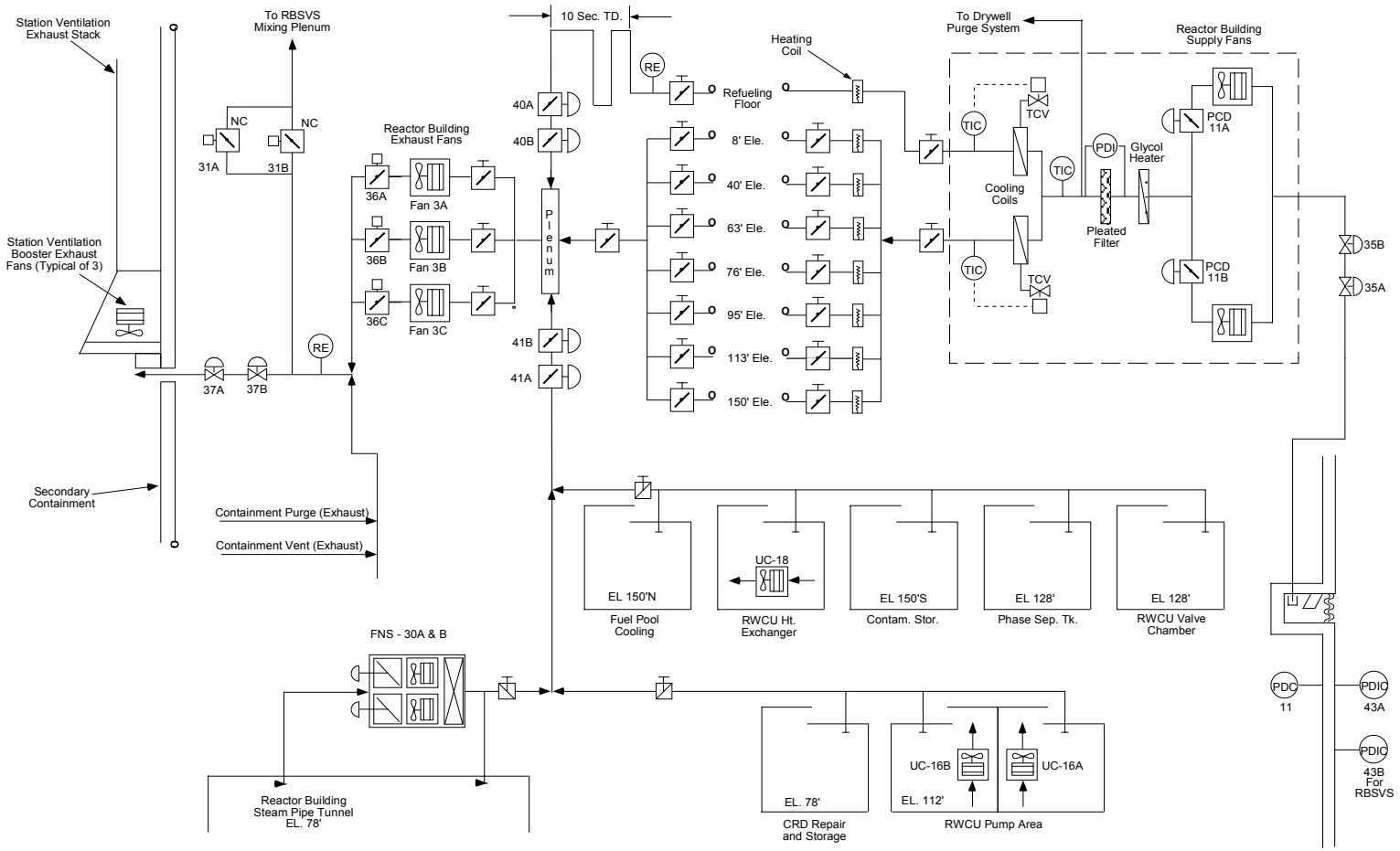


Figure 4.2.2 Reactor Building Normal Ventilation System

0396

Objectives

1. Identify the system's purposes.
2. Explain the function and operation of the following secondary containment features:
   a. Reactor building
   b. Relief panels
   c. Airlocks
   d. Reactor building normal ventilation system isolation valves and dampers
   e. Unit area coolers
   f. Primary containment purge system

Objectives

3. List the conditions that will isolate the reactor building normal ventilation system (RBNVS).
4. Explain the following flow paths:
    a. Air transfer through the RBNVS
    b. Purge vent & exhaust from primary containment
5. Explain what constitutes the secondary containment barriers.
6. Define secondary containment integrity.

Objectives

7. Summarize the response of secondary containment equipment response to accident conditions.
8. Describe how secondary containment interrelates to the following:
   a. Primary Containment System
   b. Reactor Building Standby Ventilation System
   c. Nuclear Steam Supply Shutoff System

Are there any questions?